UNIVERSITY OF BAGHDAD

COLLEGE OF MEDICINE

2027-2026

LECTURE TITLE

Organic nutrients; vitamins

Department of Family and Community Medicine

Stage Three, Semester One 2026-2027

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Date:1,4/10/2026

Learning Objectives

1. Define hidden hunger.
2. Memorize sign and symptom of vitamin A vitamin D deficiency.
3. State ways to prevent of vitamin A & vitamin D deficiency.
4. List main causes of Vitamin A and Vitamin D deficiency

Hidden hunger:

The term **“hidden hunger”** is used to describe:

- **Vitamin and mineral deficiency**.
- Resulting in health impacts may **not always be obviously visible.**
- The term was coined by WHO in 1986 and refers to the problems associated with the deficiency of **3 essential micronutrients:**

✓ **Iron.**

✓ **Vitamin A.**

✓ **Iodine.**

Vitamin A deficiency

Vitamin A divided into 2 types:

Retinoids (pre-formed vitamin/active form):

•Found only in **animal-based foods** such as eggs, fatty fish, dairy products and organ meat(beef liver, lamb liver ,chicken liver).

•**Fish liver oil (richest natural source of retinol).**

•Can be used by the **body right away.**

•**About 90 %** of ingested preformed vitamin A is absorbed.

Carotenoids/ β - carotene (pro-vitamin A):

- Found in **plant-based foods:**
  - **Red and orange fruits and vegetables** (carrot, mango, melon, apricot, and sweet potato…)
  - **Dark green leafy vegetables** (spinach, lettuce, green collard…)
- the absorption efficiency of pro-vitamin A carotenoids **varies widely**

Vitamin A deficiency(VAD): WHO defines it as “tissue concentration of vitamin A low enough to have adverse health consequences even if there is no evidence of clinical xerophthalmia”.

VAD epidemiology:

- **Leading cause of preventable blindness in children.**
- Major cause of controllable **public health and nutrition problem.**
- According to WHO statistics an estimated 250 million preschool children are vitamin A deficient.

Main Functions:

1. Vitamin A is **essential for vision** (loss of dark adaptation).
2. **Support skeletal growth** and **reproduction**.
3. Retinol and retinoic acid function as **steroid hormones**.
4. Essential for maintenance of proper **immune system.**
5. **Carotenoids** function as **antioxidants and reduce the risk of cancer.**



RDA of vitamin A

Groups	Retinol RAE(microgram)/day
➢ infants (0-6 months) ➢ infants (7-12 months)	➢ 400 ➢ 500
➢ children(1- 3 years) ➢ children (4-8 years)	➢ 300 ➢ 400
• Adolescents: ➢ boys: • 9- 13 years • 14 -18 years ➢ girls: • 9- 13 years • 14 -18 years	➢ 600 ➢ 900 ➢ 600 ➢ 700
• Adults: ➢ men: ≥ 19 years ➢ women: : ≥ 19 years	➢ 900 ➢ 700
• Pregnancy: ➢ 14 -18 years. ➢ 19 -50 years.	➢ 750 ➢ 770
• Lactation: ➢ 14 -18 years. ➢ 19 – 50 years.	➢ 1200 ➢ 1300

Risk Factors

- **Low vitamin A diet.**
- **Suboptimal breastfeeding and weaning practices:** Inappropriate breastfeeding practices, including:

✓ Early cessation of breastfeeding.

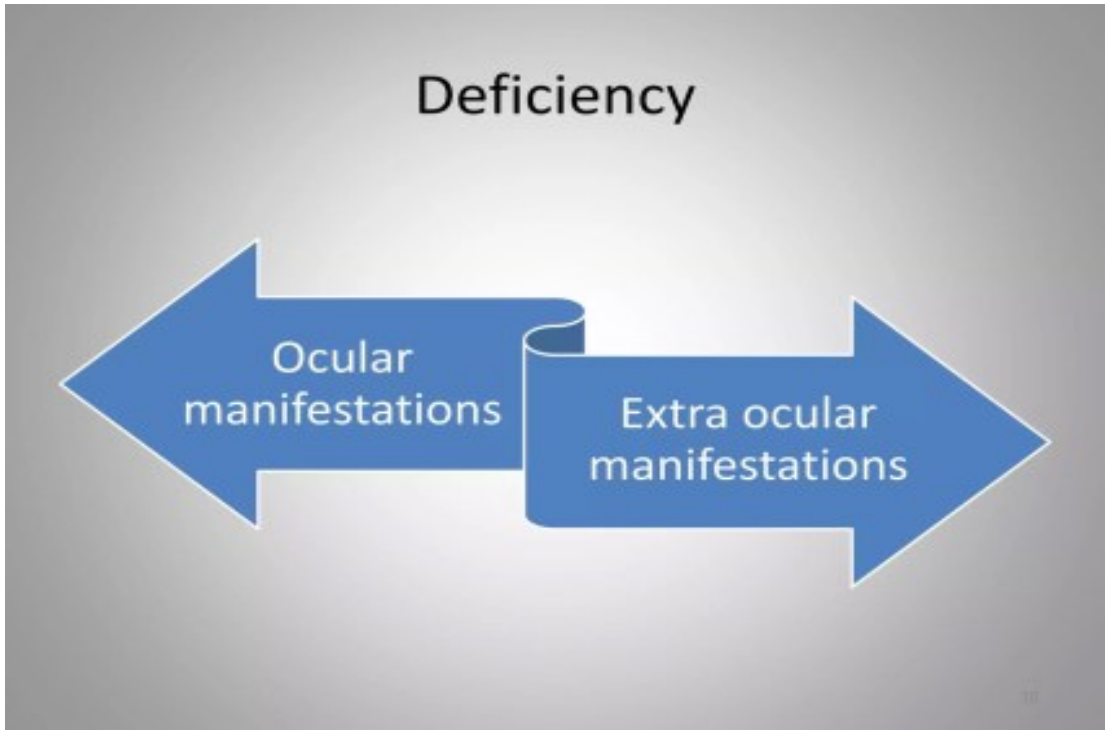
✓ Exclusive breastfeeding by mothers with insufficient vitamin A intake

✓ Infants not receiving adequate complementary foods rich in vitamin A

- **Poverty, overcrowding house and poor environmental sanitation and low educational status.**

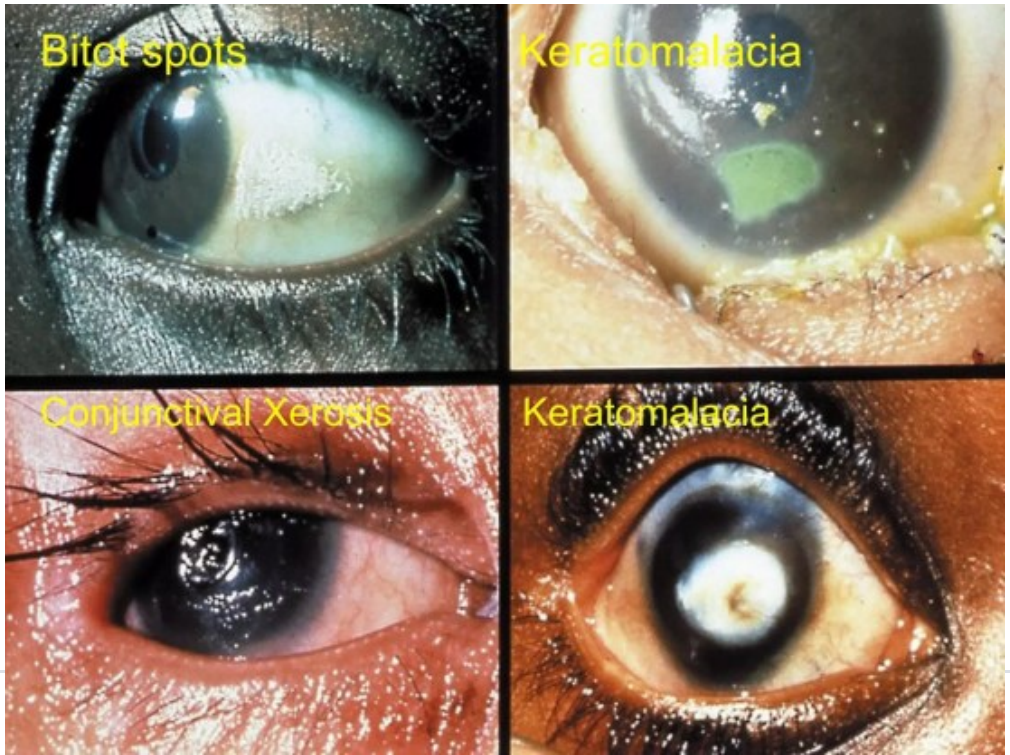
- **Infections(measles)**
- **Protein energy malnutrition.**
- **Health conditions that affect the liver**.
- **Malabsorption issues:** including short bowel syndrome, celiac disease, and bariatric surgeries (gastric bypass surgeries).
- **Vegetarians.**
- **Prolonged dietary deprivation.**
- **Chronic alcohol consumption.**

Signs and symptoms of vitamin A deficiency



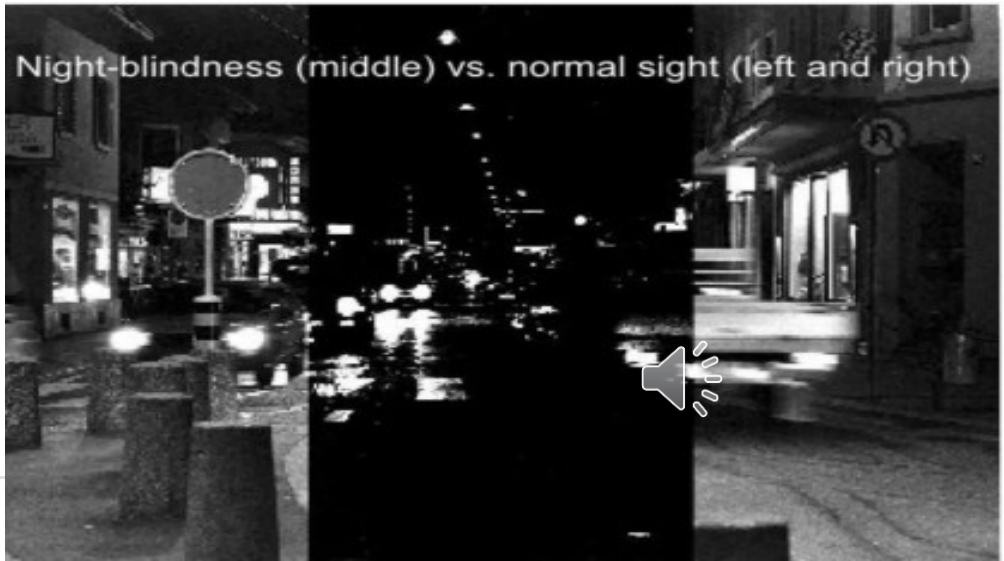
1.Ocular (specific):

- Ocular manifestations are the **most common** ones.
- **Xerophthalmia:** it refer to all the ocular manifestation of VAD, it is a serious nutritional disorder leading to preventable blindness.



A. Night blindness:

- **First symptom.**
- **Inability to see in dim light.**
- **Reversible** with vitamin A.



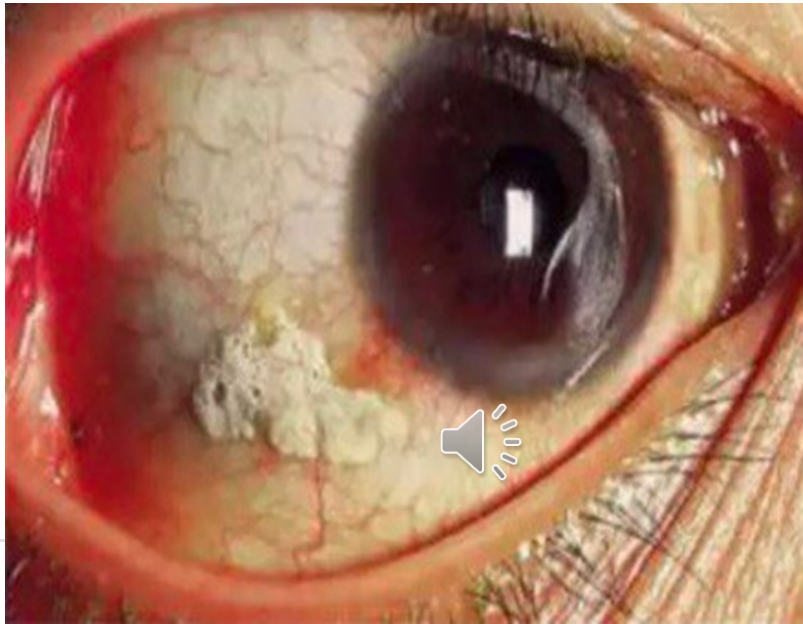
B. Conjunctival xerosis:

- **First clinical sign.**
- Cornea and tissue of eye become **dry and wrinkled.**



C. Bitot's spot:

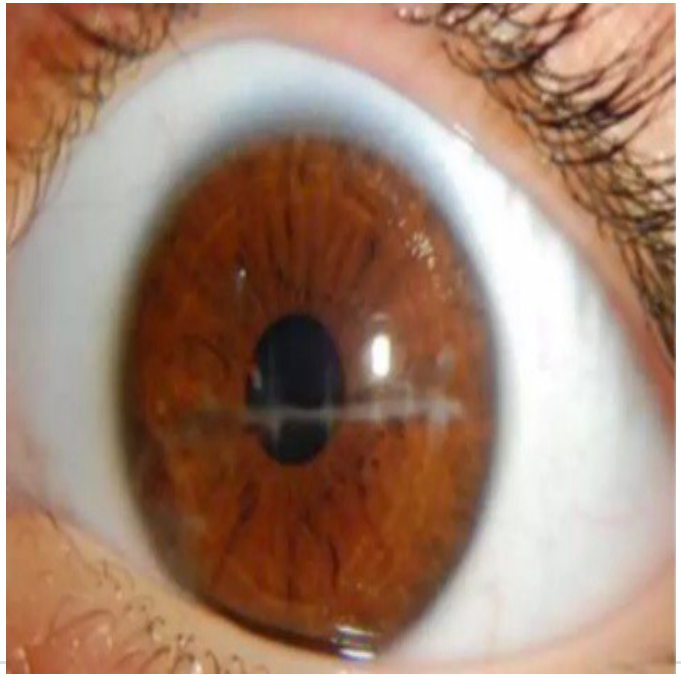
- **Triangular** pearly white to **yellowish foamy spots** on bulbar conjunctiva on either side of cornea.
- Usually **bilateral.**
- **Diagnostic in young children**



D. Corneal Xerosis:

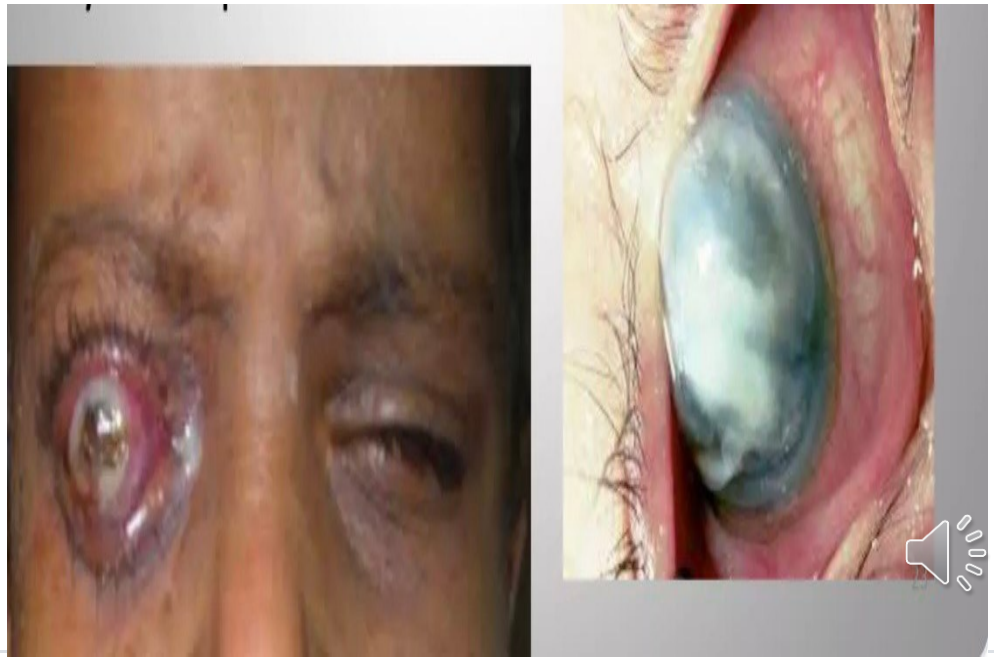
- Cornea appears **dry and dull**
- **Very serious condition → Ulcer → Scar → Affect vision.**





E. Keratomalascia:

- Liquefaction of cornea.
- A grave **medical emergency** as cornea is **soft** and may **burst open**
- **Vision is lost permanently.**



2. Extra ocular (non specific):

- **Follicular hyperkeratosis, dry brittle hair, nails, and acne**.
- Loss of appetite, weight loss
- Increased risk of anemia
- fatigue.
- **Growth retardation**
- **Low immunity** and increased vulnerability to infection
- Increased risk of certain **cancers.**

Prevalence criteria for defining vitamin A deficiency as public health significance

indicators	Cut off point
Night blindness at 24-71 months	≥ 1%
Night blindness in pregnant women	≥ 5%

Preventive and control measures of vitamin A deficiency

1. Short term: (<u>supplementation</u>) administration of large dose of vitamin A "National supplementation program with Iraqi national immunization program", **Prophylactic high doses of V.A**;

- 1st dose **100,000 IU** at **9 months** of life.
- 2nd dose **200,000 IU** at **18 months** of life.
- 3rd dose **200,000 IU** at **4-6 years** of life.

- Children with measles or protein-energy malnutrition (PEM):
  - 100,000 IU for infants <1 year
  - 200,000 IU for children ≥1 year

- Vitamin A should not be given in less than one-month intervals and all vitamin A supplementation should be recorded on each child's growth monitoring or other cards to avoid duplicate dosing.

- A high dose of Vitamin A supplementation (200,000 IU) for postpartum/lactating women(within 42 days after delivery) is no longer routinely recommended in our country.

- **High-dose** vitamin A supplementation is **contraindicated** during **pregnancy**, particularly during the **1st trimester**, due to its **teratogenic effects,** which may cause craniofacial malformations (e.g., cleft lip/palate), cardiac defects, and CNS abnormalities.
- **A well-balanced** diet supplies sufficient vitamin A.



2. Reduce or elimination of factors contributing to VAD:

- ✓ Ensure **breast feeding** and the mother should be given adequate vitamin A completed food, for bottle fed baby use **fortified formulas**.
- ✓ **Dietary modification:** Improve the diet, including more intake of foods rich in vitamin A
- ✓ Reducing frequency and severity of **PEM, respiratory infections, diarrhea and measles** with early treatment.

3. Long term prevention:

- **Foods fortification,** such as edible oils and fats, butter and milk.

- **Complete immunization** of children **<5 years**, particularly **measles vaccination**.
- **Improve sanitation and food hygiene** to reduce infections that can worsen vitamin A deficiency.
- **Nutritional education** and increasing awareness among physicians and other health personnel.
- Assessment of vitamin A deficiency through **population-based surveys**, using **clinical and biochemical indicators**.

Vitamin D deficiency

- Vitamin D is a fat-soluble vitamin.
- Unlike other vitamins, vitamin D functions like **hormone** and every single cell in the body has a receptor for it.



Epidemiology:

- Vitamin D deficiency is **a common global issue.**
- Vitamin D deficiency is **very common medical condition** in our country.

Functions of vitamin D:

1. **supporting bone health by enabling the absorption of calcium**
2. **Promoting muscle health**
3. **Modulating the immune system**
4. Eiding cell growth and differentiation.
5. **Reducing inflammation,** which helps prevent diseases such as rheumatoid arthritis and psoriasis



Risk factors for vitamin D deficiency

1. Lack or inadequate of UVB sunlight exposure:

- Air pollution
- Northern latitude.
- Occlusive garments
- Pigmented skin.
- Glass
- Elderly.
- Sunscreen
- Institutionalized, housebound, poor mobility (bedridden, wheelchair bound).

2. Breastfed infant (the vitamin D content of breast milk is low even in a vitamin D sufficient mother). Moreover, reduced maternal vitamin D stores reduce vitamin D levels transferred to the infant through placenta.

3. Prematurity ??

4. Low vitamin D diet

5. Certain **medical conditions** that affect:

- Liver disease.
- Kidney disease.

6. Certain medications:

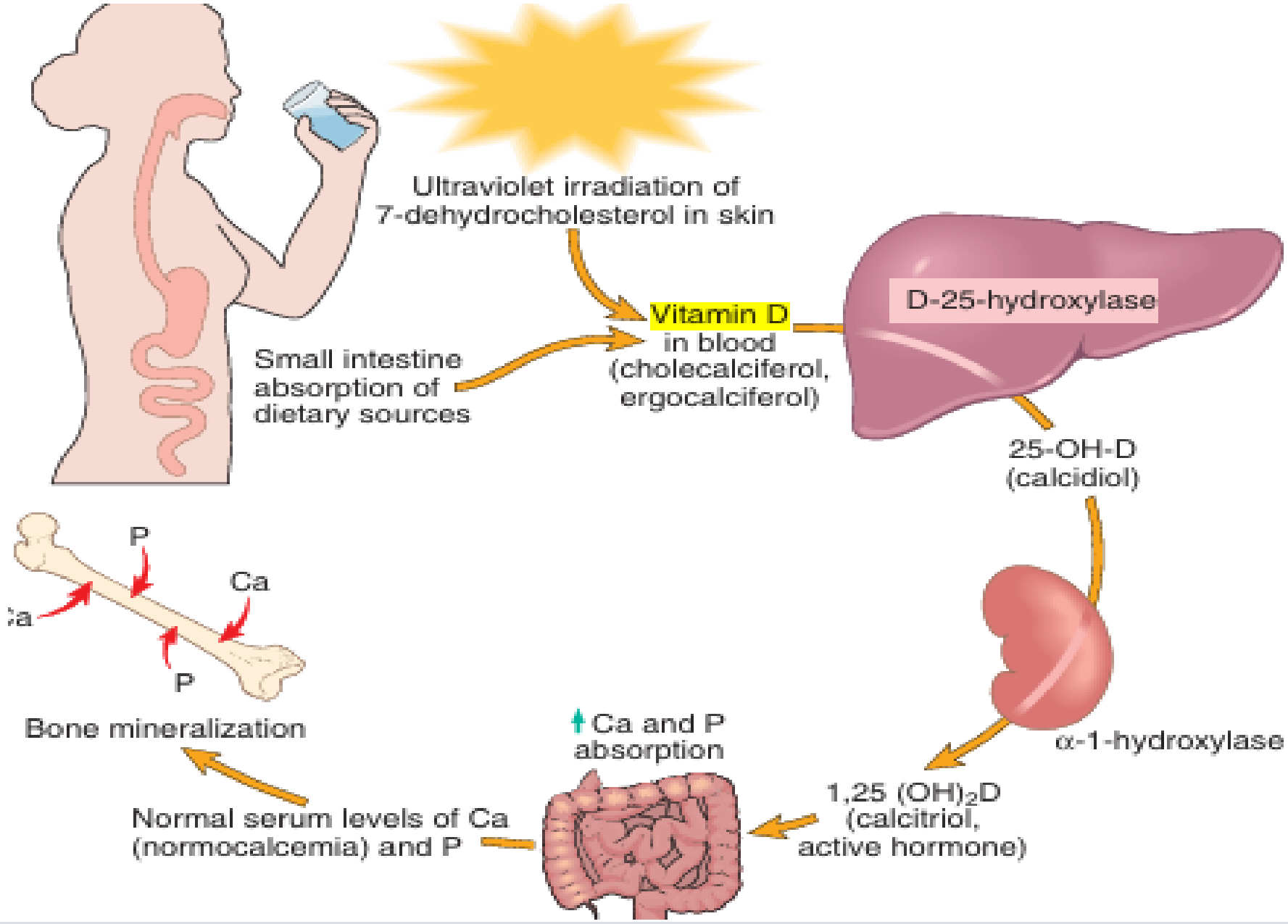
- ✓ Laxatives (long term use with high dose).
- ✓ Cholestyramine use (bile acid sequestrates).
- ✓ Orlistat (obesity medication).
- ✓ Anticonvulsants “Phenobarbital and Phenytoin”, anti TB (Rifampcin, Isoniazid), Glucocorticoid (increase breakdown/catabolism of vitamin D)



7. **Obesity** with BMI > 30 (sequestration of vitamin D in fat)

8. **Malabsorption,** including short bowel syndrome, celiac disease, inflammatory bowel diseases, and bariatric surgeries.

9. **Genetic disorder:** hereditary resistance to vitamin D, deficiency of 25-hydroxylase enzyme or 1- hydroxylase enzyme


Ultraviolet irradiation of
7-dehydrocholesterol in skin
Small intestine
absorption of
dietary sources
Vitamin D
in blood
(cholecalciferol,
ergocalciferol)
D-25-hydroxylase
25-OH-D
(calcidiol)
α-1-hydroxylase
1,25 (OH)2D
(calcitriol,
active hormone)
Ca and P
absorption
Normal serum levels of Ca
(normocalcemia) and P
Bone mineralization
P
Ca
Ca
P

Vitamin D metabolism

- UVB radiation penetrates uncovered skin and converts cutaneous 7-dehydrocholesterol to cholecalciferol or from diet cholecalciferol(D3) and ergocalciferol(D2)
- Both forms of Vitamin D are metabolized in the liver to produce initially; 25(OH) D (Calcidiol) followed by 1, 25(OH) $_2$ Vitamin D (Calcitriol) in the kidney.

Sources of vitamin D

There are 2 sources of vitamin D:

- ✓ **90% synthesized in skin** via UVB sunlight exposure.
- ✓ **10% from Dietary source:**
- ❑ The commonest dietary resources of **vitamin D3(Cholecalciferol) are animal food sources.**
- ❑ **While vitamin D2 (Ergocalciferol) is found in plant sources**



The RDA of vitamin D

Life stage	RDA
Birth to 12 months	**10 mcg (400 IU)**
Children 1-3 years	**15 mcg (600 IU)**
Teens 14–18 years	15 mcg (600 IU)
Adults 19–70 years	15 mcg (600 IU)
Adults 71 years and older	20 mcg (800 IU)
Pregnant and breastfeeding teens and women	15 mcg (600 IU)

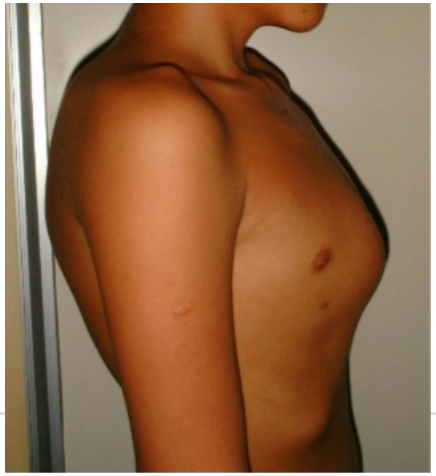
Signs and symptoms of vitamin D deficiency in children

Severe vitamin D deficiency in children causes **rickets** with features include:





- ✓ **Bowed legs,** often the most obvious sign.
- ✓ Skeletal deformities **(kyphoscoliosis, Pigeon chest).**
- ✓ Beaded ribs **(rachitic rosary)**





- ✓ **Craniotabes** (softening of the skull bones).
- ✓ Bone pain and muscle weakness.
- ✓ Muscle cramps or **seizures**(hypocalcemia)
- ✓ **Widening of the distal radius and ulna on X-Ray.**

Signs and symptoms of vitamin D deficiency in adults

Lack of vitamin D isn't quite as obvious in adults:

- ✓ **Osteomalacia,** occurs in women of childbearing age
- ✓ **Osteoporosis and fracture.**
- ✓ Fatigue, **falls** and impaired physical function.
- ✓ **Bone pain** in lower back pain, pelvis or lower extremities.
- ✓ **Muscle weakness;** especially proximal muscle weakness
- ✓ Mood changes, like **depression.**
- ✓ **Increased susceptibility to infection.**

Vitamin D status definitions according to the Endocrine Society Clinical Practice Guideline

Cut-off value of vitamin D	25(OH) D(ng/ml)
Deficiency	< 20*
Insufficiency	21-29
Sufficiency	30-100
toxicity	>100**

* A serum 25(OH)D level <10 ng/mL is considered severe vitamin D deficiency.

** Excessive vitamin D supplementation can lead to hypercalcemia, kidney stones, and vascular calcification.

Prevention and control measures

1. Sunlight Exposure: Ultraviolet UV-B radiation.

- ✓ Mid day sun (10 am- 3pm) for average of 5-15 minute/day at least twice a week depends on the skin color.
- ✓ Avoid excessive sun exposure and sunburn (erythema), as UV radiation can damage the skin and increase the risk of skin.

2. Dietary Intake of vitamin D–rich foods:

- **Vitamin D3(Cholecalciferol)** mainly animal sources:
  - ✓ Fatty fish (salmon, mackerel, sardines)
  - ✓ Cod liver oil
  - ✓ Egg yolks
  - ✓ Fortified milk
- **Vitamin D2 (Ergocalciferol) in plant sources**:
  - ✓ **Mushrooms** exposed to UV light (provide little vitamin D).

3. Supplementation:

- Particularly important for:
  - **Infants** (especially breastfed)
  - **Elderly** (less skin synthesis due to atrophied subcutaneous fat).
  - **Pregnant/lactating women**
  - People with **limited sun exposure** or **dark skin**
- Typical **prophylactic doses:**
  - Infants: **400 IU/day**
  - Adults: **600–800 IU/day**
- **Treatment doses** may be prescribed if deficiency is present.

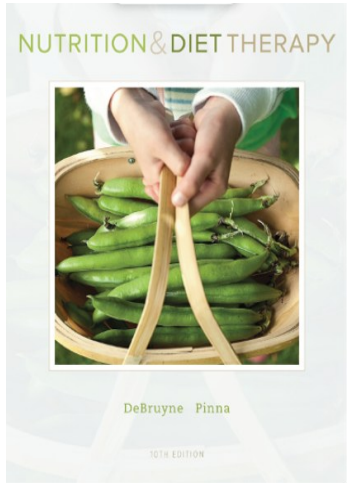
4. Managing Risk Factors:

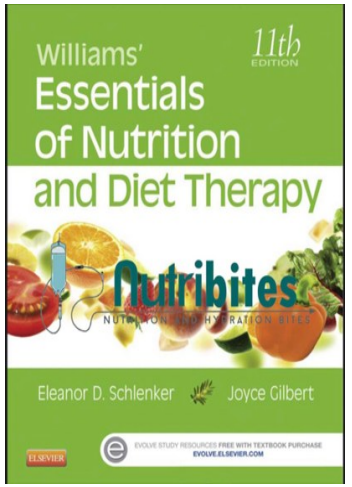
- **Malabsorption disorders** (celiac, IBD, cystic fibrosis) → may need higher doses or injectable forms.
- **Obesity** → vitamin D is sequestered in fat; may require higher intake.**??**
- **Medications** (e.g., cholestyramine, anticonvulsants) → monitor and supplement accordingly.
- **Critically ill neonate:** Measuring serum 25-OH vitamin D level, to detect potential deficiency or insufficiency early

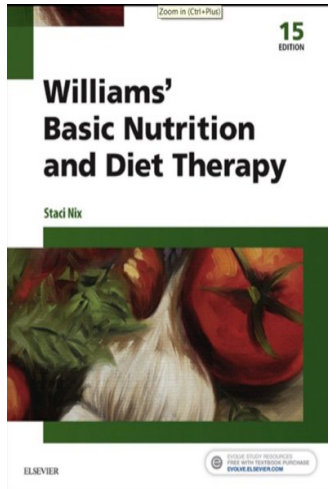
In summery:

- Hidden hunger define as vitamin and mineral deficiency may not always be obviously visible.(iron, vitamin A and iodine)
- Ocular manifestations are the most common in vitamin A def. ….Xerophthalmia
- National supplementation program of vitamin A in Iraq: 1st dose 100,000 IU at 9 months of life. 2nd dose 200,000 IU at 18 months of life. 3rd dose 200,000 IU at 4-6 years of life.
- 90% of vitamin D sources is synthesized in skin via UVB sunlight exposure.
- Features of vitamin D deficiency in children rickets(Bowed legs, Skeletal deformities, rachitic rosary, Craniotabes….)
- Excess vitamin A is contraindicated in pregnancy particularly during 1st trimester, because of risk of teratogenicity

References







1. Nutrition and Diet Therapy 10th edition

2. Williams' Basic Nutrition and Diet Therapy 15 edition

3. Williams' Essentials of Nutrition and Diet Therapy 11th edition

QUESTIONS & DISCUSSION



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